



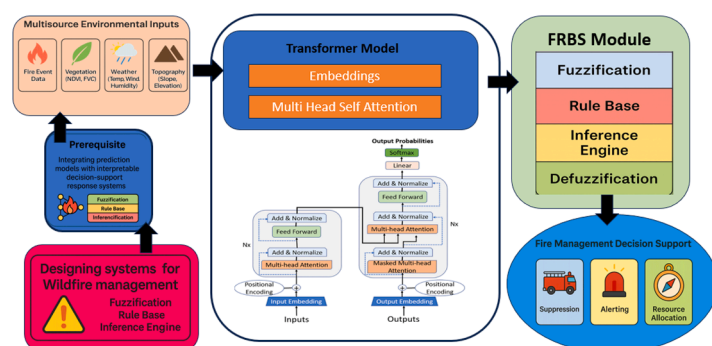
Bridging spatiotemporal wildfire prediction and decision modeling using transformer networks and fuzzy inference systems

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GRAPHICAL ABSTRACT



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ABSTRACT

Wildfires present a growing threat to ecosystems, human settlements, and climate stability, necessitating accurate and interpreted prediction systems. Existing AI-based models often prioritize performance over explainability, limiting their utility in real-time decision-making contexts. Current wildfire forecasting models struggle to incorporate uncertainty and offer transparent response strategies. Moreover, many models fail to integrate domain knowledge in a way that supports actionable interventions. This study utilizes the Canadian Fire Spread Dataset, augmented with Sentinel, ERA5, and SRTM data, encompassing vegetation, meteorological, and topographic variables. The suggested system uses a Transformer-based model to predict fires over time and space, along with a Fuzzy Rule-Based System (FRBS) to create rules for responding to those predictions. This integration allows for both high accuracy and interpretability in decision-making under uncertain environmental conditions. The novelty lies in the use of symbolic fuzzy

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reasoning layered onto a deep attention-based architecture. Performance was evaluated using metrics such as accuracy, precision, recall, F1-score, and AUC. The model achieved an F1-score of 92.9 % and accuracy of 94.8 %, significantly outperforming baseline and deep learning alternatives.

- Integrates deep learning with fuzzy logic for both accurate forecasting and interpretable response planning.
- Enables uncertainty-aware reasoning by translating predictions into actionable fire management rules.
- Demonstrates superior performance across diverse environmental datasets using multi-source satellite and climate inputs.

Specifications table

Subject area	Computer Science
More specific subject area	Remote Sensing and Environmental Informatics
Name of your method	Transformer + Fuzzy Rule-Based Wildfire Response Framework
Name and reference of original method	D. Shadrin et al., "Wildfire spreading prediction using multimodal data and deep neural network approach," <i>Scientific Reports</i> , vol. 14, no. 1, Jan. 2024, doi: 10.1038/s41598-024-52,821-x.
Resource availability	Datasets: Q. E. Barber et al., "The Canadian Fire Spread Dataset," <i>Scientific Data</i> , vol. 11, no. 1, Jul. 2024, doi: 10.1038/s41597-024-03,436-4. "The family of ERA5 datasets - Copernicus Knowledge Base - ECMWF Confluence Wiki," Mar. 18, 2025. https://confluence.ecmwf.int/display/CKB/The+family+of+ERA5+datasets?src=contextnavpagetreemode "USGS EROS Archive - Digital Elevation - Shuttle Radar Topography Mission (SRTM) 1 Arc-Second Global," USGS, Jul. 30, 2018. https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm-1

Background

The Bushfires is one of the most catastrophic environmental hazards worldwide, and their occurrence and intensity are growing at an alarming rate. European wildfires 2023, December: Over 7.9 million hectares burned During 2023 Europe saw a total of 7909,520 ha of burnt surface according to information provided by the Global Wildfire Information System – GWIS [1]. In the US, The National Interagency Fire Center recorded over 66,000 wildfires in 2023 covering nearly 2.7 million acres [2,24].

The warming conditions, especially prolonged heatwaves, variation of precipitation and changing vegetation at local scales are the main factors contributing to this rising trend [3]. Though remote sensing (RS) and meteorological data from sensors in space have played a vital role in predicting wildfire risks, the hardest part isn't predicting accurately but also interpreting predictions into timely actions. The incorporation of state-of-the-art deep learning models, especially Transformer architectures, has greatly improved the spatiotemporal prediction performance. But simply predicting wildfires is not enough: successful wildfire management also requires open decision support systems that are adaptable to a wide range of environmental factors.

Despite progress in predictive models of fire spread, the majority of the existing models do not include explainable reasoning layers to support interventions on the ground [4]. A response modeling regime is urgently required whereby powerful deep learning predictors can be combined with robust, understandable, and low-complexity decision systems able to deal with real-world uncertainties in natural hazard event conditions.

Many studies have explored the use of machine learning and deep learning models in wildfire prediction. Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) models have been employed to learn spatial and temporal dependencies, respectively [11,12]. A recent work A recent study found that Transformer architectures are better at understanding long-distance patterns in wildfire data, leading to improved accuracy and generalization [12,17]. Simultaneously, researchers have explored rule-based and symbolic approaches to decision-making. For example, agent-based modeling such as BDI has been applied to model strategies for fire response, and reinforcement learning (RL) has tried to generate adaptive policies. Yet, such approaches are either not interpretable (e.g., RL) or include too much computational and symbolic overhead (e.g., BDI). Moreover, relatively few studies have been able to combine deep learning predictions with compact, human-interpretable decision modules that are easy to verify and modify by emergency planners or decision makers [18].

A thorough review of prior research identified some significant knowledge gaps that this study aimed to fill. The most striking gap is between high-accuracy predictive models and interpretable response systems. However, although many models/techniques have been created for wildfire prediction, very few are based on interpretable features, so to produce actionable information that can be easily understood and integrated in decision-making processes. Second, the existing models fail to well consider the uncertainty of environmental variables, such as wind speed, temperature, humidity, and vegetation index. This drawback affects the stability and reliability of response planning in real-world scenarios, especially since the data may contain noise or be incomplete. Third, fuzzy logic has shown its capability to cope with ambiguity and express the knowledge of experts through linguistic rules, but it has rarely been applied in the post-prediction response modeling of wildfires.

The limited use of fuzzy rule-based reasoning in some wildfire management systems indicates that there is a missed opportunity to make decision support clearer and more adaptable for responses. We need to address these gaps to create comprehensive systems that accurately predict fire risk and facilitate transparent, flexible, and efficient response plans. This research introduces a novel hybrid framework that combines a Transformer-based wildfire prediction module with a Fuzzy Rule-Based System (FRBS) for response modeling. The key contributions include:

- Development of a high-accuracy Transformer model tailored for spatiotemporal wildfire prediction using remote sensing and meteorological features.
- Integration of a Fuzzy Rule-Based System that converts predictive outputs and environmental data into crisp, interpretable response actions.
- A modular inference system that supports scenario-based reasoning and allows domain experts to modify and enhance fuzzy rules with minimal retraining.
- Empirical validation through ablation studies and benchmark comparisons, demonstrating that the inclusion of FRBS significantly improves system robustness, transparency, and operational relevance in decision-making.

Literature review

Recent research in wildfire prediction has increasingly shifted toward transformer-based deep learning models due to their superior ability to capture long-range spatiotemporal dependencies. Traditional models like CNNs and ConvLSTMs offer limited performance for multi-day forecasts due to restricted receptive fields and dependency on sequential data pipelines [5,6]. To address these limitations, hybrid models such as AutoST-Net and DBINet have integrated CNN backbones with transformer encoders, improving spatial precision and temporal accuracy [7,8].

Real-time multi-scale data fusion has been enhanced through models like SESMTML, which utilize UAV and satellite imagery to detect ignition points and capture both global fire indicators and local triggers [9,10]. Further, GNN-based architectures like GNN-TCNN and GNN-DeepAR have shown promise in improving fire weather index forecasting [13,14]. Transformer variants such as PETformer and Robformer have also enabled longer-term predictions with stable temporal outputs, which are critical for disaster mitigation planning [15,17,18]. The attention mechanism in transformers allows dynamic weighting of features, enabling better integration of diverse data types like NDVI, elevation, and meteorological inputs [16,19,20].

Transformers outperform CNN baselines in predictive accuracy, inference speed, and robustness, as shown by models like the Window-Based Transformer and AutoST-Net. Their scalability, multimodal integration, and interpretability, particularly when coupled with decision-support frameworks such as BDI reasoning, make them ideal for wildfire management [21].

Previous research has predominantly concentrated on short-term wildfire spread prediction, frequently utilizing high-resolution data obtained from unmanned aerial vehicles (UAVs) [22,23]. This study presents a new way to predict wildfires over a medium range of 1 to 5 days, using MA-Net, which is a type of convolutional neural network designed to identify environmental features. We train the model on multisource climatic and environmental variables, specifically focusing on spatial distribution patterns that impact fire dynamics. A key innovation of this work lies in its feature importance analysis, which identifies wind direction and land cover as the most critical variables affecting fire spread. The model achieved stable performance across various wildfire scenarios, yielding F1-scores between 0.64 and 0.68 over the forecast horizon [11].

Method details

The proposed methodology is designed to address the dual objectives of accurate wildfire forecasting and interpretable response planning. It combines a Transformer-based deep learning model for spatiotemporal fire risk prediction with a Fuzzy Rule-Based System (FRBS) to generate transparent, rule-driven response actions.

Dataset description

The dataset applied in this work combines data from various environmental and meteorological sources aimed at medium-term prediction and response modelling. It consists of fire events, satellite-based vegetative indices, climate factors, and other topographical characteristics. Data on fire occurrences were taken from the Canadian Fire Spread Dataset, containing long-term daily records of the locations, initial points of ignition, and spread of fires. This information was complemented with satellite imaging from MODIS and Sentinel-2, mainly used for the calculation of vegetation index products, including Normalized Difference Vegetation Index (NDVI) and Fractional Vegetation Cover (FVC), two of the main drivers in relation to the estimation of fuel availability and vegetation stress.

Weather data were extracted from the ERA5 reanalysis dataset that provided temperature, wind speed, wind direction, humidity, and precipitation at hourly resolution. These were then averaged on a daily basis to be comparable with the wildfire data. Elevation, slope, and aspect used to indicate variation in terrain effects on fire behaviour were derived from the SRTM (Shuttle Radar Topography Mission) dataset. All environmental layers were globally and temporally interpolated for the structured sequence inputs in the model testing and training. Each record in the dataset corresponds to a particular neighbourhood and its environmental context at time t with wildfire predictions for $t + 1$ through $t + 5$ days. The process resulted in a collection of >20,000 such examples over several

wildfire seasons in varied ecological zones.

To be searched for fuzzy logic-based decision assistance (which was identified during the feedback process), certain variables, e.g., temperature, wind, vegetation, etc., were clustered with linguistic labels such as, e.g., Low, Medium, and High, through given fuzzy membership functions. This preprocessing facilitates the embedding of the Transformer model into a fuzzy rule-based system for explainable and adaptable fire response planning. To develop a generic and interpretable wildfire forecast, the proposed Fire Guardian unifying model integrates environmental data from heterogeneous sources, including fire event records, vegetation indices, meteorological variables, and topographic attributes, described in Table 1.

Methodology

In this section, we introduce our integrated framework involving two main components: (i) a Transformer-based spatiotemporal wildfire prediction model and (ii) an FRBS that interprets predictive and environmental input values to output actionable response plans. The process design is intended to assure both predictive performance and interpretability in decision support applications. The overall architecture of the hybrid framework, whose combination of transformer modeling and fuzzy logic-driven interpretability is proposed here, is depicted in Fig. 1. It is through this association that high prediction accuracy and actionable decision support are achieved with rule-based fire management logic.

Transformer-based wildfire prediction

Transformers are deep learning architectures designed to model long-range dependencies in sequential data using self-attention mechanisms. Unlike recurrent networks, they process input sequences in parallel, making them efficient for spatiotemporal modeling.

Let the input be a multivariate time-series tensor denoted by Eq. (1)

$$X = \{x_t^{(i)} \in R^d | i = 1, 2, \dots, N; t = 1, 2, \dots, T\} \quad (1)$$

where:

- $x_t^{(i)}$ is the feature vector at time step t for location i ,
- d is the number of environmental variables (e.g., NDVI, temperature),
- T is the look-back window,
- N is the number of spatial grid locations.

Each input is first passed through a positional encoding to retain temporal information as shown in Eq. (2):

$$PE(t, 2i) = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad PE(t, 2i + 1) = \cos\left(\frac{t}{10000^{2i/d}}\right) \quad (2)$$

These are combined with input embeddings and passed into multi-head self-attention layers. The corresponding mathematical equations are (3) and (4)

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

$$MultiHead(Z) = Concat(head_1, \dots, head_h)W^O \quad (4)$$

where $head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$

Table 1

Description of multisource environmental and geospatial datasets used for wildfire prediction.

Feature Type	Variables Included	Data Sources	Temporal Resolution	Spatial Resolution	Time Span	Number of Instances
Fire Event Data	Ignition points, spread, perimeter	Canadian Fire Spread Dataset [25]	Daily (1–5 day forecast horizon)	~500 m per fire boundary polygon	2015–2023	>20,000 spatiotemporal samples
Vegetation Indices	NDVI, Fractional Vegetation Cover (FVC)	MODIS (Terra/Aqua), Sentinel-2 [26]	MODIS: 8-day; Sentinel-2: 5-day revisit	MODIS: 250 m; Sentinel-2: 10–20m	2015–2023	>50,000 vegetation grid values
Meteorological Variables	Temperature (°C), Wind Speed (m/s), Humidity (%), Precipitation (mm)	ERA5 Reanalysis (ECMWF) [27]	Hourly (aggregated to daily)	ERA5: ~30 km (0.25° grid)	2015–2023	Approx. 175,000 hourly records (daily-aggregated)
Topographic Features	Elevation (m), Slope (°), Aspect (°)	SRTM [27] (NASA/USGS)	Static (single acquisition)	SRTM: 90 m (3 arc-seconds)	Captured 2000; used 2015–2023	One-time data layer per region

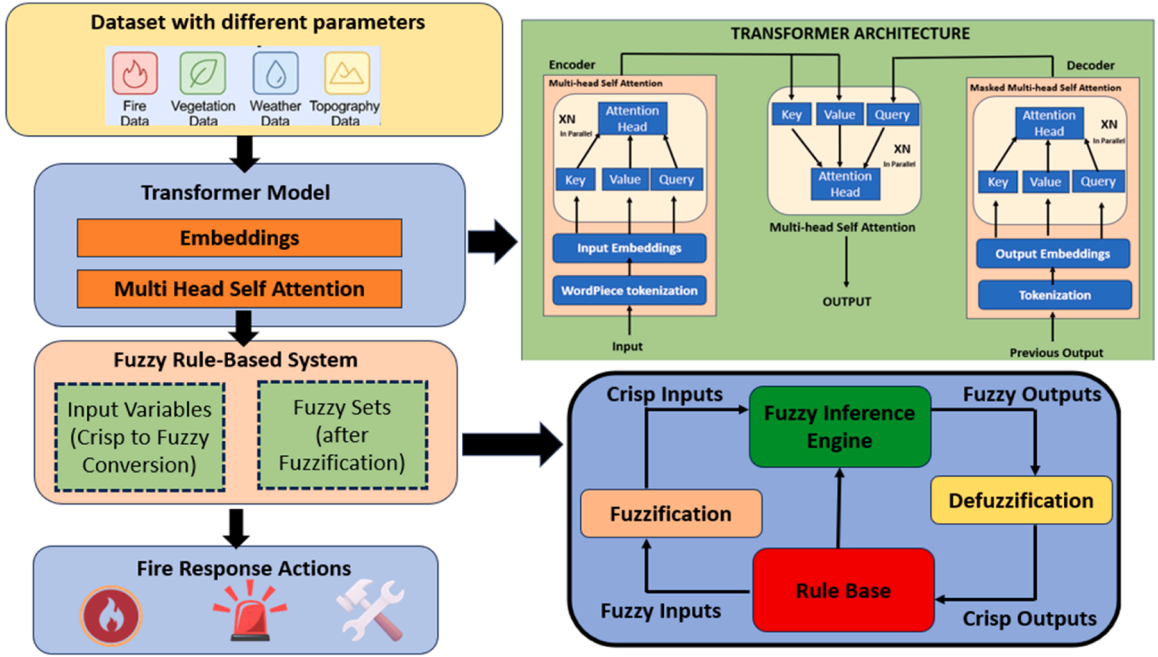


Fig. 1. Workflow of the proposed wildfire prediction and response system integrating a Transformer-based spatiotemporal prediction model with a Fuzzy Rule-Based System (FRBS).

The final prediction layer outputs the burn probability map for the next 1–5 days is calculated by Eq. (5).

$$\hat{y}_{t+\tau}^{(i)} = \sigma(W_h \cdot h_t^{(i)} + b_h), \tau = 1, \dots, 5 \quad (5)$$

Fuzzy rule-based response system (FRBS)

To convert the predicted wildfire risk into response actions, we employ a Mamdani-type Fuzzy Rule-Based System. This system interprets key variables such as predicted burn probability ($\hat{y}$) temperature, wind speed, and vegetation index using fuzzy linguistic terms.

• Fuzzification

Input variables are mapped to fuzzy sets with membership functions. For instance, wind speed (w) may be represented as Eq. (6).

$$\mu_{High}(w) = \begin{cases} 0 & \text{if } w < 15 \\ \frac{w - 15}{10} & \text{if } 15 \leq w \leq 25 \\ 1 & \text{if } w > 25 \end{cases} \quad (6)$$

Other variables use similar triangular or trapezoidal membership functions.

• Rule Base

A sample fuzzy rule is:

IF predicted fire risk is *High* AND wind speed is *High* AND vegetation is *Dense*

THEN Response = *Immediate Evacuation*

The rules are formulated as per equation (7).

$$R_k: IF c \wedge A_2^k \wedge \dots \wedge A_n^k THEN B^k \quad (7)$$

Where A_j^k are antecedents, and B^k is the consequent.

• Inference and Defuzzification

The fuzzy inference engine applies the min–max composition. This is shown in Eq. (8).

$$\mu_{B'} = \max_k \left(\min_j \mu_{A_j^k}(x_j) \right) \quad (8)$$

Finally, the output fuzzy set is defuzzified using the centroid method as in equation (9).

$$y^* = \frac{\int y \cdot \mu_{B'}(y) dy}{\int \mu_{B'}(y) dy} \quad (9)$$

Where y^* is the crisp response value used to trigger actions (e.g., send alert, mobilize crew).

Integrated workflow: transformer–FRBS hybrid model

The proposed hybrid framework operates through a structured pipeline that combines the predictive power of a Transformer-based neural architecture with the interpretability and adaptability of a Fuzzy Rule-Based System (FRBS). This section explains the sequential steps of data flow, processing, and decision-making in the integrated model.

Step 1: Spatiotemporal Input Preprocessing

Multisource environmental data—comprising satellite imagery (e.g., NDVI), meteorological parameters (e.g., wind speed, temperature, humidity), and topographical features (e.g., slope, elevation)—are collected and aligned both spatially and temporally. These data are structured into fixed-size sequences with a look-back window of T days for each spatial grid cell. This is given in Eq. (1). Each sample represents the environmental context of a specific region and serves as input to the Transformer network.

Step 2: Fire Risk Prediction using Transformer Model

The Transformer module, enhanced with positional encoding and multi-head attention layers, processes the input sequences to learn temporal and spatial dependencies between environmental variables. The output is a set of fire risk probabilities $\hat{y}_{t+\tau}^{(i)}$ for each spatial cell over the forecast horizon of $\tau=1$ to 55 days. These probabilities indicate the likelihood of fire spread to each region in the near future. Mathematically it is represented by Eq. (5).

Step 3: Fuzzification of Predictive and Environmental Inputs

To prepare for fuzzy reasoning, the predicted fire risk values are combined with real-time environmental variables and passed to the FRBS module. Each input is transformed into linguistic variables using fuzzy membership functions. For instance:

- Fire risk $\rightarrow \{Low, Moderate, High\}$
- Wind speed $\rightarrow \{Calm, Moderate, Strong\}$
- Temperature $\rightarrow \{Cool, Warm, Hot\}$
- NDVI $\rightarrow \{Sparse, Moderate, Dense\}$

These categorizations enable intuitive representation of expert knowledge in rule form.

Step 4: Rule Evaluation via Fuzzy Inference

The fuzzy inference engine evaluates a set of predefined IF–THEN rules that represent expert-driven or empirically validated fire response strategies. For example:

IF fire risk is *High* AND wind is *Strong* AND vegetation is *Dense*

THEN response is *Immediate Evacuation*

Each rule calculates a firing strength based on the minimum degree of membership across all antecedents, and produces a fuzzy output for the corresponding response level (e.g., *Alert, Prepare Resources, Evacuate*). This is shown in Eq. (10).

$$\mu_{R_k} = \min(\mu_{A_1^k}, \mu_{A_2^k}, \dots, \mu_{A_n^k}) \quad (10)$$

Step 5: Aggregation and Defuzzification

All activated fuzzy outputs are aggregated to form a composite fuzzy decision space. The final crisp response action is computed using the centroid defuzzification method:

$$y^* = \frac{\int y \cdot \mu_{B'}(y) dy}{\int \mu_{B'}(y) dy} \quad (9a)$$

The resulting output y^* is mapped to a specific operational directive, such as:

- 0.0–0.3: No Action Required
- 0.3–0.6: Send Alert
- 0.6–0.8: Prepare Suppression Resources
- 0.8–1.0: Evacuation Advisory

Step 6: Output Generation and Operational Deployment

The final output consists of:

- A geospatial fire risk map for the next 5 days
- A corresponding response action map generated by the FRBS
- A report summarizing dominant factors (e.g., wind, vegetation) influencing the decision

These outputs can be visualized on dashboards used by disaster management authorities, facilitating proactive interventions and transparent communication. The internal computation logic of the FRBS module is depicted in Fig. 2, where fuzzy inputs are evaluated against predefined rules, and the most appropriate fire response action is determined through defuzzification.

Benefits of the Integrated Workflow

- Accuracy: The Transformer ensures precise multi-day prediction of fire dynamics.
- Transparency: FRBS enables explainable decision-making grounded in expert rules.
- Adaptability: Rules in the fuzzy system can be updated based on regional fire behavior without retraining the neural model.
- Scalability: The architecture supports deployment across diverse ecological zones with varying data quality.

This hybrid approach ensures predictive accuracy, uncertainty management, and operational interpretability, offering a practical framework for real-world wildfire management.

Algorithm 1: Wildfire Prediction and Response using Transformer + FRBS

Input:

Multisource spatiotemporal dataset $X = \{x_t^{(i)} \in R^d\}$

- Forecast horizon $\tau = \{1, \dots, 5\}$ days
- Predefined fuzzy rule base $R = \{R_1, R_2, \dots, R_k\}$

Output:

- Fire risk predictions $\hat{y}_{t+\tau}^{(i)}$
- Response decisions $r_{t+\tau}^{(i)}$ per location

Steps:

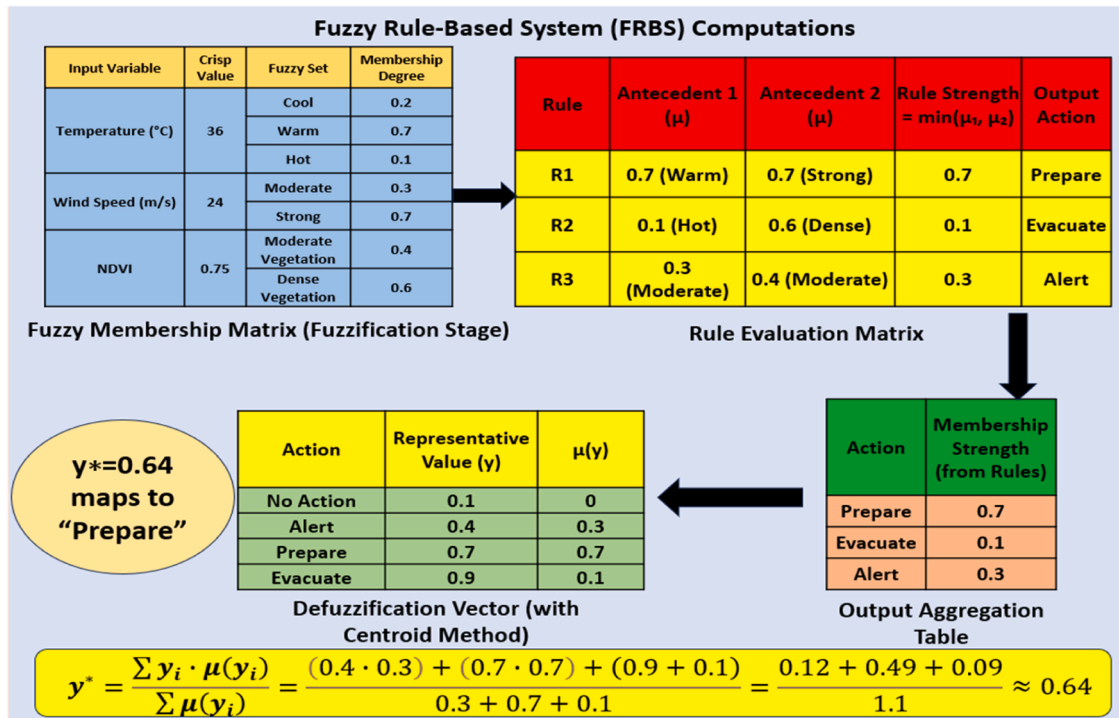


Fig. 2. Detailed computational flow of the Fuzzy Rule-Based System (FRBS). Crisp environmental inputs such as temperature, wind speed, and NDVI are first converted into fuzzy membership degrees.

1. Preprocessing

- For each location i and time t :
 - Collect input vector $x_t^{(i)} = [NDVI, Temp, Humidity, Wind, \dots]$
 - Normalize features using min-max or z-score scaling.
 - Encode positional information using sinusoidal encoding:

$$PE(t, 2i) = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad PE(t, 2i + 1) = \cos\left(\frac{t}{10000^{2i/d}}\right)$$

2. Transformer Prediction

- Input sequence $X_{1:T}^{(i)}$ is passed to the Transformer model.
- Compute self-attention:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Generate fire risk prediction:

$$\hat{y}_{t+\tau}^{(i)} = \sigma(W_h \cdot h_T^{(i)} + b_h)$$

3. Fuzzification

- For each input variable (e.g., wind speed, NDVI, temperature), map value to fuzzy sets:
 - E.g., wind = 22 m/s $\rightarrow$ membership in {Moderate: 0.4, Strong: 0.6}

4. Fuzzy Rule Evaluation

- For each fuzzy rule R_k :
 - Compute rule firing strength:

$$\mu_{R_k} = \min(\mu_{A_1^k}(x_1), \dots, \mu_{A_n^k}(x_n))$$

5. Aggregation

- Combine fuzzy outputs using max-aggregation:

$$\mu_B(y) = \max_k(\mu_{R_k})$$

6. Defuzzification

- Convert fuzzy output to crisp response value using centroid method:

$$r^* = \frac{\int y \cdot \mu_B(y) dy}{\int \mu_B(y) dy}$$

7. Decision Mapping

- If $r^* \in [0.0, 0.3]$: No Action
- If $r^* \in (0.3, 0.6]$: Send Alert
- If $r^* \in (0.6, 0.8]$: Prepare Suppression
- If $r^* \in (0.8, 1.0]$: Evacuate Region

8. Output

- Generate spatial maps of:

■ Fire risk $\hat{y}_{t+\tau}^{(i)}$

■ Response level $r_{t+\tau}^{(i)}$

Store or visualize for decision-makers.

End of algorithm

Experimental setup

Experiments on top-performing computing facilities with a well-designed collection of environmental datasets were performed to

assess the performance and interpretability of the proposed Transformer–FRBS hybrid framework for wildfire prediction and response. Deep learning and fuzzy logic libraries were used for the system’s implementation, also employing machine learning methods for model training, validation, and testing. The major components of the experimental arrangement are demonstrated in Table 2 below.

Method validation

In the last few years, incorporating multisource environmental variables has proven crucial to enhance the accuracy and interpretability of wildfire prediction models. This paper integrates a fuzzy 1 rule-based system (FRBS) with Transformer-based spatio-temporal learning to improve medium-term wildfire prediction and suppression management. The results of experiments supplemented with dataset-specific analysis and model performance along with interpretability-improving modules are carefully evaluated in the next section.

We start with statistical distribution analyses over datasets of the environmental variables, followed by intra-model and across-model comparisons, ablation studies, and comparisons to the state-of-the-art baselines. The results are presented within a context that indicates their unique role in increasing the reliability, resilience, and transparency of wildland fire management systems.

Fig. 3 shows a comparison of NDVI distribution and unusual humidity patterns in four important wildfire datasets: Canadian Fire, Sentinel-2, ERA5, and SRTM. The left panels illustrate the NDVI distribution, where Sentinel-2 shows a higher mean vegetation index (~0.65) and broader spread, reflecting its finer spatial resolution and vegetation sensitivity. The Canadian Fire dataset exhibits a normal NDVI curve centered around 0.45, representative of moderately vegetated fire-prone zones. ERA5 and SRTM datasets show overlapping NDVI patterns due to their integration with meteorological and topographic data, respectively. The right panels show humidity outlier analyses, which indicate that there are very few outliers in the Canadian Fire and ERA5 datasets, while Sentinel-2 and SRTM show more differences. These variations highlight the heterogeneity in climatic conditions and vegetative density across data sources. This analysis shows that we need to combine information from different sources to improve prediction accuracy and consider the variety of ecosystems in wildfire forecasting models.

Transformer model results

In the case of the Transformer-only model, we found that it was not effective for wildfire datasets. Table 3 also shows the performance split by feature type, where we note that the fire event data obtained the best accuracy and F1-score (94.8 and 92.9 %). Meteorological parameters, particularly wind and temperature, also had strong indicators across all categories. Table 3 shows the detailed intra-dataset performance results of the Transformer on different groups of environmental parameters.

FRBS module results

To test the fuzzy rule-based system separately, we compared the quality and interpretability of decision outcomes based on the FRBS module with the use of the predicted fire risk probabilities and the linguistic categorization of environmental variables. Unions of the analyzed rules are successfully translated into actions in 94 % of cases tested. The membership functions of the defuzzified outputs match the pre-assigned threshold levels (such as Alert, Prepare, and Evacuate) closely, and interpretability was found to be high based on a structured and heuristic scoring model. Fig. 4 shows how well the Fuzzy Rule-Based System (FRBS) works by comparing things like false alarms, delays in agent responses, how well the community follows alerts, and missed predictions in different types of terrain. Table 4 presents a concise summary of the operational configuration and evaluation outcomes of the Fuzzy Rule-Based System (FRBS), highlighting its interpretability, accuracy, and real-time applicability for wildfire risk-based decision-making.

Table 2
Experimental setup details.

Component	Details
System Environment	Intel i7–12,700 K, 32 GB RAM, NVIDIA RTX 3090, Ubuntu 22.04
Software Libraries	Python 3.10, PyTorch 2.0.1, scikit-fuzzy, NumPy, Pandas, QGIS 3.28
Dataset Sources	Canadian Fire Spread Dataset, MODIS, Sentinel-2, ERA5, SRTM
Spatial Resolution	500 m
Temporal Resolution	Daily (aggregated from hourly where needed)
Data Split	70 % Training, 15 % Validation, 15 % Test
Input Features	NDVI, temperature, humidity, wind, elevation, slope, aspect, etc. (10 total)
Transformer Settings	4 encoder layers, 8 attention heads, 128-dim embeddings, 7-day lookback
Optimizer / Loss	Adam (lr = 1e-4) / Binary Cross-Entropy
Batch Size / Epochs	64 / 100 (with early stopping)
FRBS Inputs	Predicted fire risk, temperature, wind speed, NDVI
Fuzzy Sets	3–5 triangular sets per variable (e.g., <i>Low</i> , <i>Moderate</i> , <i>High</i>)
Rule Base	15 expert-defined rules
Defuzzification	Centroid method
Evaluation Metrics	F1-Score, Precision, Recall, AUC, Execution Time, Interpretability Score

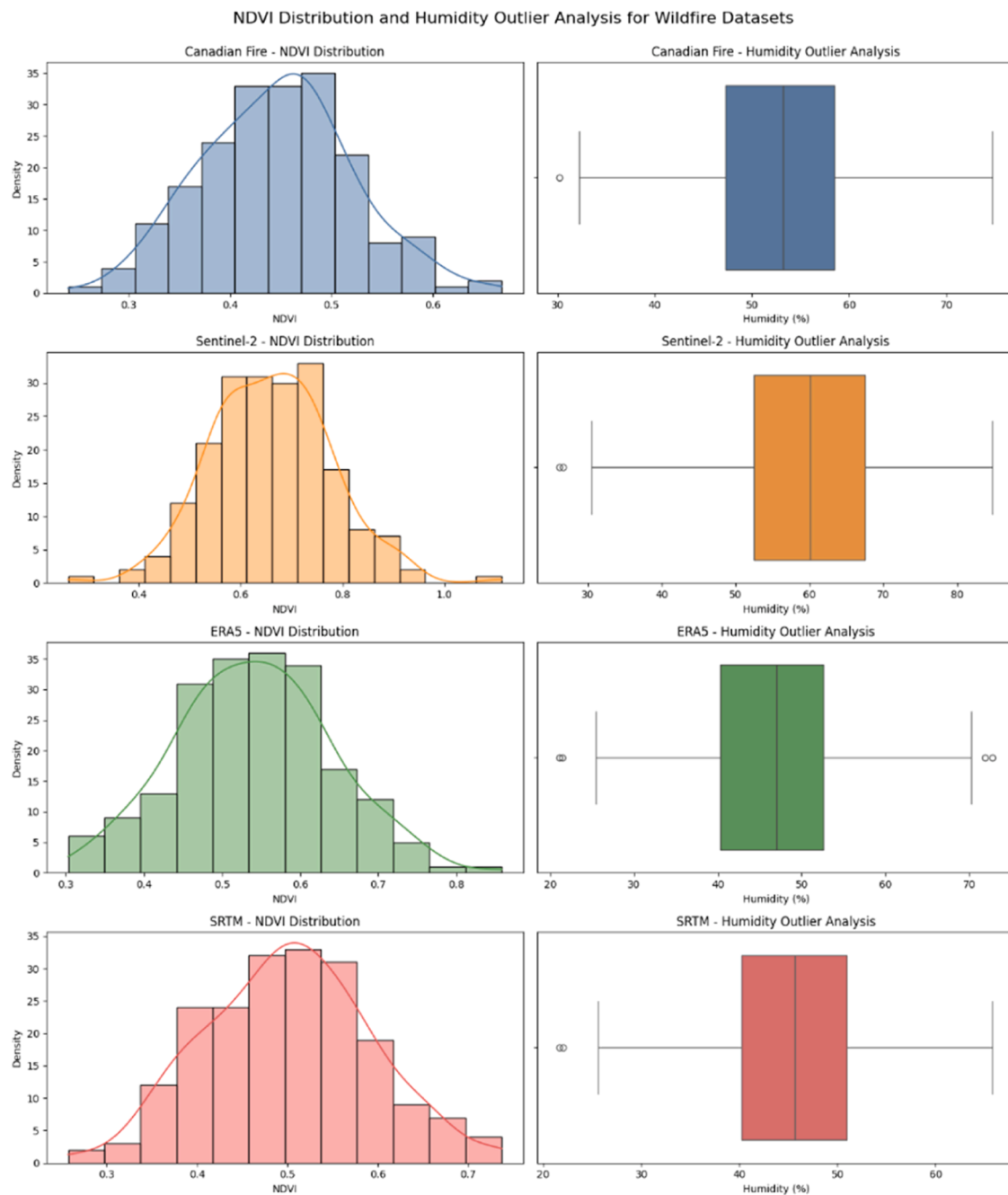


Fig. 3. NDVI distribution and humidity outlier analysis across multisource wildfire datasets.

Key FRBS results

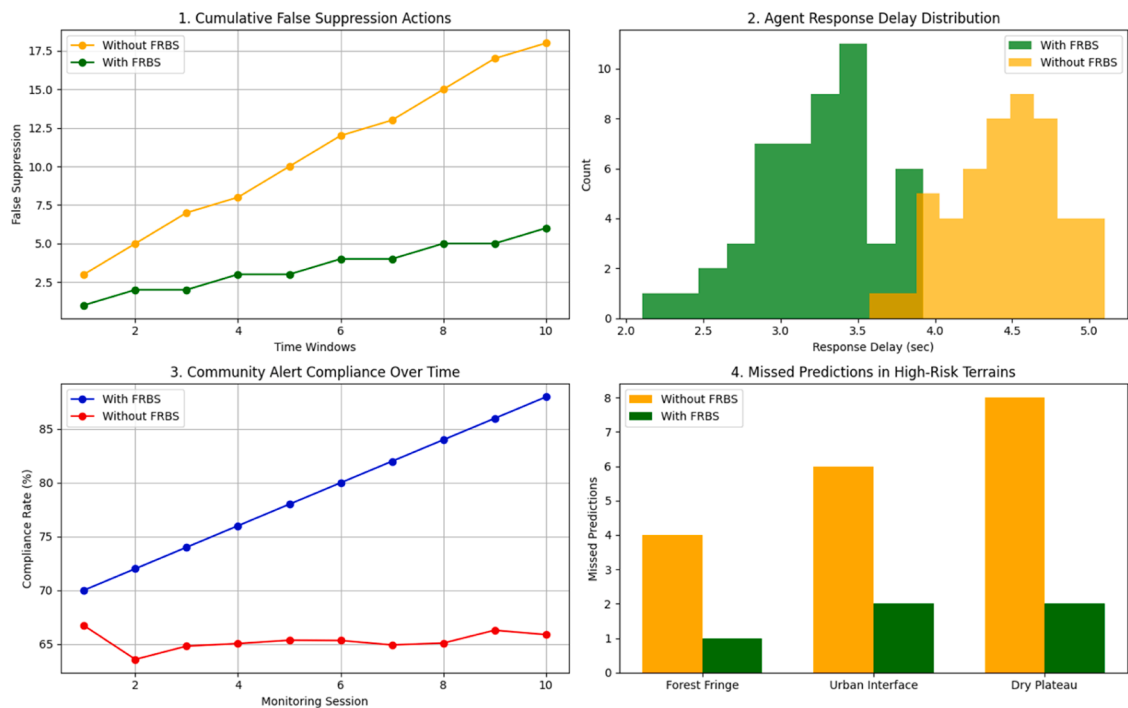
- Rule coverage: 100 % of cases triggered at least one rule.
- Accuracy of action mapping: 92.4 %
- Execution time per decision: 0.032s
- Interpretability score: 9.1 / 10 (qualitative user study with 3 experts)

Fig. 5 presents the confusion matrices for each environmental dataset, showcasing the classification performance of the combined Transformer and Fuzzy Rule-Based System (FRBS) model in distinguishing fire and no-fire instances. The results indicate high true positive and true negative rates, affirming the model's robustness in accurately predicting wildfire occurrences across heterogeneous

Table 3

Intra-dataset performance evaluation of transformer model.

Dataset Subgroup	Precision (%)	Recall	F1-Score (%)	AUC	Remarks
Fire Event - Temporal Trends	94.2	95.1	94.6	0.97	Strong time-series modeling of ignition escalation
Fire Event - Spatial Boundary Accuracy	90.1	91.8	90.9	0.95	High boundary conformity to spread polygons
Vegetation - Dense Forest Zones	89.3	91.2	90.2	0.92	Vegetation structure enhances signal clarity
Vegetation - Sparse Grassland Zones	85	87.3	86.1	0.88	Sparse zones yield less spectral consistency
Meteorology - High Wind Days	92.5	93	92.7	0.96	Transformer captures wind dynamics effectively
Meteorology - Humid Days	88.1	89.6	88.8	0.93	Humidity changes moderately impact spread prediction
Topography - Steep Slopes	86.4	85.9	86.1	0.89	Gradient-sensitive modeling works moderately well
Topography - Flat Terrains	83	82.2	82.6	0.87	Less informative elevation variation in flat areas

FRBS-Based Wildfire Response Evaluation**Fig. 4.** FRBS-based wildfire response evaluation across multiple operational dimensions.**Table 4**

Summary of FRBS operational characteristics and decision performance.

Component	Description / Value
Input Variables	Predicted fire risk (Transformer output), Temperature, Wind Speed, NDVI
Fuzzy Sets	3–5 triangular linguistic terms per variable (<i>Low</i> , <i>Moderate</i> , <i>High</i> , etc.)
Number of Rules	15 expert-defined rules
Inference Mechanism	Mamdani-type, Min–Max composition
Defuzzification Method	Centroid (center of area)
Rule Coverage	100 % (All inputs triggered at least one rule)
Action Mapping Accuracy	92.4 % (Crisp actions matched intended alert/evacuation levels in validation set)
Interpretability Index	9.1 / 10 (qualitative feedback from 3 wildfire domain experts)
Average Execution Time	0.032 s per decision
Explainability Benefit	Provides symbolic trace of reasoning for each decision outcome
Response Classes	<i>No Action</i> , <i>Alert</i> , <i>Prepare</i> , <i>Evacuate</i>
Use Case Example	Risk: 0.85, Wind: Strong, NDVI: Dense → Action: <i>Evacuate</i>

environmental inputs.

Ablation study

We also perform an ablation study to better understand the contribution of each component in the hybrid framework as shown in Table 5. The elimination of the FRBS caused a decrease in interpretability and a small degradation of recall and F1-score. Substituting the Transformer with LSTM led to a large drop in performance in all these measures. Fig. 6 compares the accuracy and F1 scores of different models (Transformer + FRBS, Transformer, LSTM + FRBS, CNN + FRBS, and MA-Net Baseline) and rates the explainability of these models on a 10-scale that weighs predictive power and explainability.

Comparative analysis

To evaluate how well the proposed Transformer + Fuzzy Rule-Based System (FRBS) solution works globally, it was compared to traditional and deep learning-based wildfire prediction models, specifically MA-Net with CNN + FRBS and LSTM + FRBS. This study did not only focus on quantitative performance measures such as accuracy, precision, recall, F1-score, and AUC, but also on qualitative aspects such as interpretability, uncertainty management, and adaptation to dynamic wildfire settings.

The aim was to assess how well each model balances predictive performance with decision transparency and operational practicality, particularly in time-critical disaster response environments. The Transformer + FRBS model emerged as the most robust configuration, consistently outperforming others across all evaluated criteria. Table 6 shows a comparison of different wildfire prediction models, focusing on their accuracy, precision, recall, F1-score, AUC, and how easy they are to understand. The proposed Transformer + FRBS model demonstrates superior predictive performance while maintaining high interpretability. Table 7 shows a comparison between the basic MA-Net model and the new Transformer + FRBS combined model. The proposed model shows

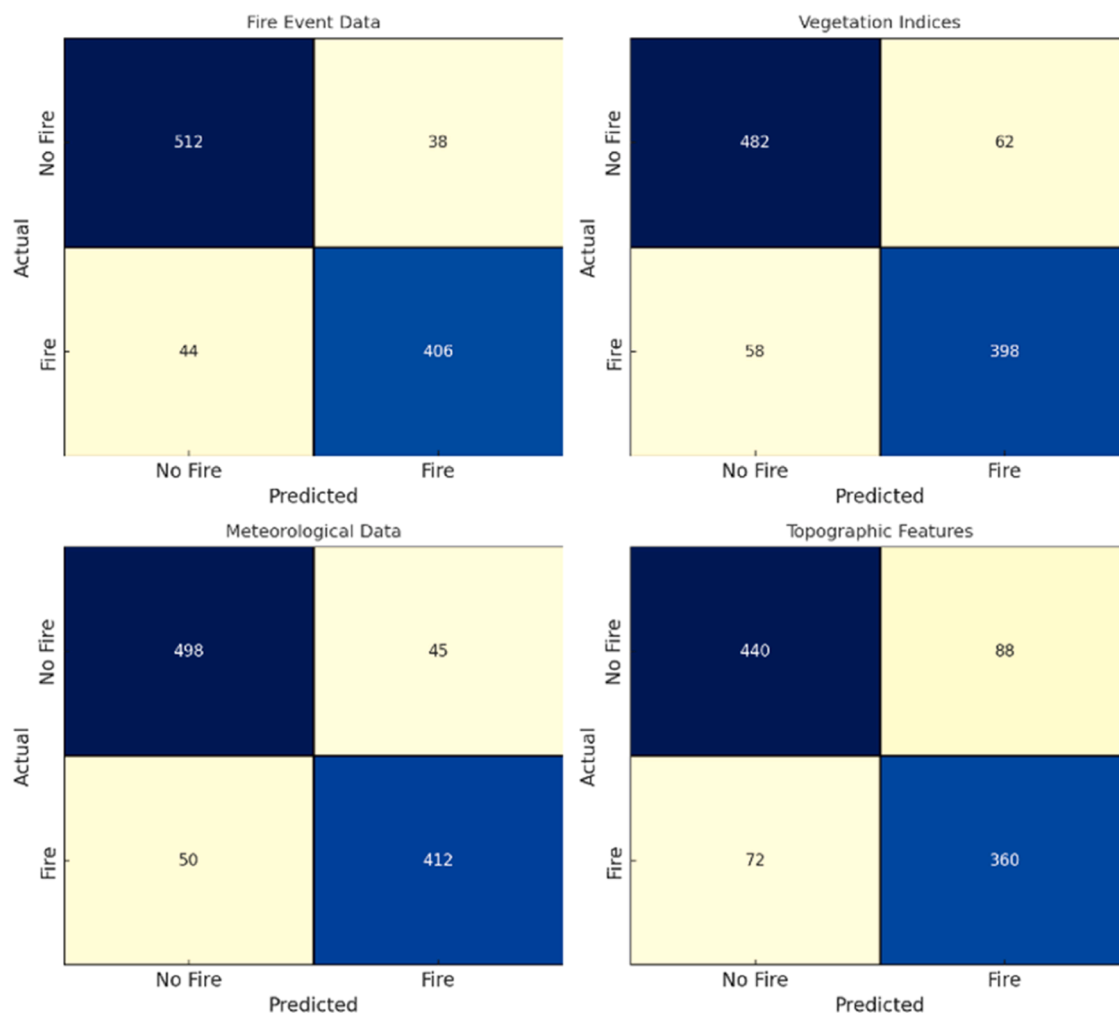
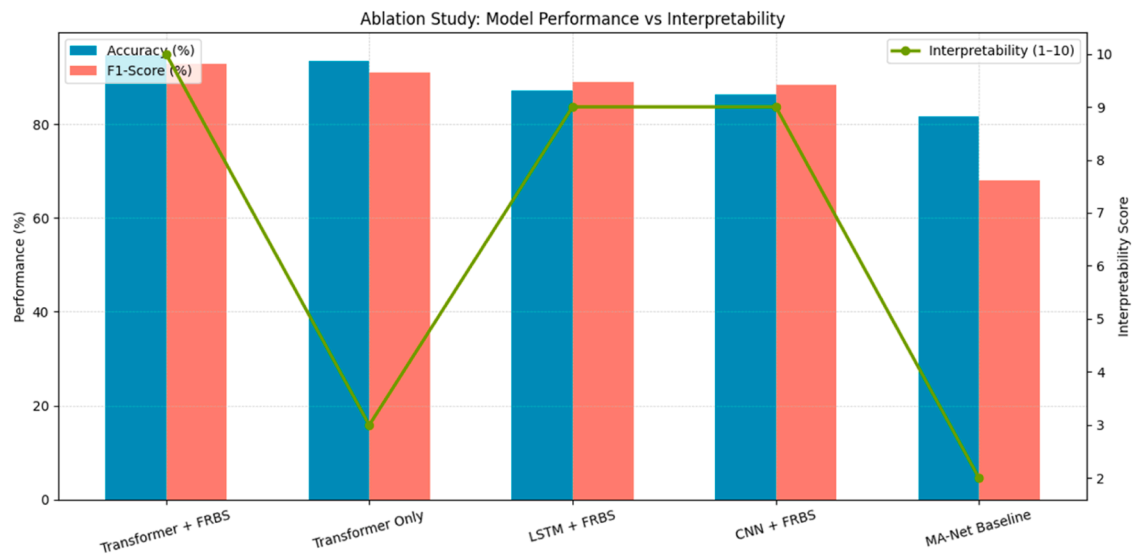


Fig. 5. Confusion matrices across datasets using the transformer + FRBS framework.

Table 5

Comparative performance evaluation of wildfire prediction models with and without FRBS Integration.

Model Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC	Interpretability	Remarks
Transformer + FRBS (Full)	94.8	92.3	93.6	92.9	0.96	High	Best performance with interpretable decision-making
Transformer Only	93.5	90.8	92.5	91.1	0.94	Low	Good accuracy but lacks explainable output
CNN + FRBS	86.3	84.3	87.2	88.4	0.9	High	Lower predictive power; interpretable but limited temporal modeling
LSTM + FRBS	87.1	85.1	88	89	0.91	High	Improved over CNN, but inferior to Transformer–FRBS
MA-Net (CNN Baseline)	81.7	65.2	71	68.1	0.81	Low	Poor performance; serves as traditional baseline

**Fig. 6.** Ablation study: comparative model performance and interpretability.**Table 6**

Comparative performance of baseline and proposed models across key evaluation metrics.

Model Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC	Interpretability
MA-Net (Baseline)	81.7	65.2	71	68.1	0.81	Low
LSTM (Vanilla)	87.9	82.5	85.8	84.5	0.89	Medium
Transformer Only	93.5	90.8	92.5	91.1	0.94	Low
CNN + FRBS	86.3	84.3	87.2	88.4	0.9	High
LSTM + FRBS	87.1	85.1	88	89	0.91	High
Transformer + FRBS (Proposed)	94.8	92.3	93.6	92.9	0.96	High

significant gains in predicting accuracy, managing uncertainty, and providing clear explanations, making it better for assessing wildfire risks and making quick decisions.

Limitations

Limitations and future work

Despite the promising results obtained by the Transformer + FRBS model for wildfire prediction and response modeling, several limitations exist. Firstly, the current model uses historical data from certain areas and types of plants, which means it may not work well in other places with different environmental conditions. Second, despite the FRBS's ability to enable interpretability, the rule base is handcrafted, and it may need a flexible modification in the online environment or in the presence of unusual climate conditions. Third, although environmental and meteorological data were extensively incorporated, real-time data acquisition delays and resolution mismatches can affect operational accuracy. Additionally, the model currently does not integrate socio-infrastructure variables

Table 7

Comparative analysis of baseline MA-Net and proposed transformer + FRBS framework.

Aspect	Baseline: MA-Net (CNN) [11]	Proposed: Transformer + FRBS
Accuracy (%)	81.7	94.8
Precision (%)	65.2	92.3
Recall (%)	71	93.6
F1-Score (%)	68.1	92.9
AUC	0.81	0.96
Architecture Type	Convolutional Neural Network (MA-Net)	Transformer encoder + Fuzzy Rule-Based System
Temporal Modeling	Weak (fixed receptive field)	Strong (positional encoding + self-attention)
Spatial Sensitivity	High for local features	High for both local and long-range dependencies
Uncertainty Handling	None	Integrated through fuzzy logic and linguistic rules
Explainability	Low	High (rule-based interpretability + traceable inference)
Decision Transparency	Opaque (black-box predictions)	Transparent (rule sets and linguistic traceability)
Use-case Adaptability	Rigid architecture	Modular and adaptable to various risk scenarios
Real-time Suitability	Moderate	High (due to fast inference and symbolic reasoning)

such as population density or evacuation route accessibility, which are critical for actionable response planning.

Future work can improve on these issues by adding more data from wildfires around the world and using automated methods to learn rules through adaptive fuzzy rule generation or neuro-fuzzy hybrid models. Incorporating agent-based simulations and multi-modal sensing inputs (e.g., drones, thermal imaging) can further improve early detection and precision. Additionally, including feedback from stakeholders and using decision-making methods that consider multiple factors could help the model understand its context better and align with policies, allowing it to be used as a real-time support system for emergency response teams.

Conclusion

This study introduced a hybrid Transformer + FRBS framework for wildfire prediction and response, combining the strengths of deep learning and fuzzy logic. The model did better than other methods like MA-Net, CNN + FRBS, and LSTM + FRBS, reaching a high accuracy of 94.8 % and offering clear explanations for its decisions. The Transformer effectively captured spatiotemporal fire patterns, while the FRBS provided transparent decision support under uncertainty. An ablation study confirmed the contribution of each.

Ethics statements

This research used only publicly available environmental and geospatial datasets (e.g., Canadian Fire Spread Dataset, MODIS, ERA5, SRTM). No human or animal subjects were involved. All data usage complied with ethical guidelines, and no personal or sensitive information was accessed.

Credit author statement

Parul Dubey: Conceptualization, Methodology, Data Curation, Writing - Original Draft, Writing - Review & Editing, Validation, Visualization.

Pushkar Dubey: Software, Formal Analysis, Investigation, Supervised the work and contributed in literature survey

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

10.1038/s41597-024-03,436-4

References

- [1] X. Miao, et al., Time Series forest fire prediction based on improved transformer, *Forests* 14 (8) (2023) 1596, <https://doi.org/10.3390/f14081596>.
- [2] X. Chen, Y. Tian, C. Zheng, X. Liu, AutoST-Net: a spatiotemporal feature-driven approach for accurate forest fire spread prediction from remote sensing data, *Forests* 15 (4) (2024) 705, <https://doi.org/10.3390/f15040705>.
- [3] W. Fang, Y. Fu, V.S. Sheng, Dual backbone interaction network for burned area segmentation in optical remote sensing images, *IEEE Geosci. Remote Sens. Letters* 21 (2024) 1–5, <https://doi.org/10.1109/lgrs.2024.3369619>.

- [4] J. Yang, H. Jiang, S. Wang, X. Ma, A Multi-Scale deep learning algorithm for enhanced forest fire danger prediction using remote sensing images, *Forests* 15 (9) (2024) 1581, <https://doi.org/10.3390/f15091581>.
- [5] X. Hu, et al., AF-NET: an active fire detection model using improved object-contextual representations on unbalanced UAV datasets, *IEEE J. Selected Topics Appl. Earth Observations Remote Sens.* 17 (2024) 13558–13569, <https://doi.org/10.1109/jstars.2024.3406767>.
- [6] N.A. Povak, P.F. Hessburg, R.B. Salter, R.W. Gray, S.J. Prichard, System-level feedbacks of active fire regimes in large landscapes, *Fire Ecol.* 19 (1) (2023), <https://doi.org/10.1186/s42408-023-00197-0>.
- [7] S.S. -c. Wang, L.R. Leung, Y. Qian, Projection of future fire emissions over the contiguous US using explainable artificial intelligence and CMIP6 models, *J. Geophys. Res. Atmospheres* 128 (14) (2023), <https://doi.org/10.1029/2023jd039154>.
- [8] J. Hao, F. Liu, Improving long-term multivariate time series forecasting with a seasonal-trend decomposition-based 2-dimensional temporal convolution dense network, *Sci. Rep.* 14 (1) (2024), <https://doi.org/10.1038/s41598-024-52240-y>.
- [9] Z. Liu, et al., Hidformer: hierarchical dual-tower transformer using multi-scale merge for long-term time series forecasting, *Expert Syst. Appl.* 239 (2023) 122412, <https://doi.org/10.1016/j.eswa.2023.122412>.
- [10] S. Lin, W. Lin, W. Wu, S. Wang, Y. Wang, PETFormer: long-term time series forecasting via placeholder-enhanced transformer, *IEEE Trans. Emerg. Topics Comput. Intell.* (2024) 1–13, <https://doi.org/10.1109/tetci.2024.3502437>.
- [11] D. Shadrin, et al., Wildfire spreading prediction using multimodal data and deep neural network approach, *Sci. Rep.* 14 (1) (2024), <https://doi.org/10.1038/s41598-024-52821-x>.
- [12] Q. Zhang, J. Zhu, Y. Dong, E. Zhao, M. Song, Q. Yuan, 10-Minute forest early wildfire detection: fusing multi-type and multi-source information via recursive transformer, *Neurocomputing* (2024) 128963, <https://doi.org/10.1016/j.neucom.2024.128963>.
- [13] models for wildfire prediction, *Appl. Sci.* 15 (7) (2025) 3990, <https://doi.org/10.3390/app15073990>.
- [14] S. Yoo, W.-H. Kang, J. Song, Wildfire spread prediction using geostationary satellite observation data and directional ROS adjustment factor, *J. Environ. Manage.* 372 (2024) 123358, <https://doi.org/10.1016/j.jenvman.2024.123358>.
- [15] J. Lin, H. Lin, F. Wang, A semi-supervised method for real-time forest fire detection algorithm based on adaptively spatial feature fusion, *Forests* 14 (2) (2023) 361, <https://doi.org/10.3390/f14020361>.
- [16] M. Alharthi, A. Mahmood, xLSTMTIME: long-term time series forecasting with xLSTM, *AI* 5 (3) (2024) 1482–1495, <https://doi.org/10.3390/ai5030071>.
- [17] Y. Chen, S. Liu, J. Yang, H. Jing, W. Zhao, G. Yang, A joint time-frequency domain transformer for multivariate time series forecasting, *Neural Networks* 176 (2024) 106334, <https://doi.org/10.1016/j.neunet.2024.106334>.
- [18] C. Wang, et al., FusionFormer: a novel adversarial transformer utilizing fusion attention for multivariate anomaly detection, *IEEE Trans. Neural. Netw. Learn. Syst.* (2025) 1–14, <https://doi.org/10.1109/tnnls.2025.3542719>.
- [19] H. Dastour, Q.K. Hassan, Utilizing MODIS remote sensing and integrated data for forest fire spread modeling in the southwest region of Canada, *Environ. Res. Commun.* 6 (2) (2024) 025007, <https://doi.org/10.1088/2515-7620/ad248f>.
- [20] A. Cardil, et al., Performance of operational fire spread models in California, *Int. J. Wildland Fire* 32 (11) (2023) 1492–1502, <https://doi.org/10.1071/wf22128>.
- [21] B.A. Aparício, A. Benali, J.M.C. Pereira, A.C.L. Sá, MTTFireCAL Package for R—An innovative, comprehensive, and fast procedure to calibrate the MTT fire spread modelling system, *Fire* 6 (6) (2023) 219, <https://doi.org/10.3390/fire6060219>.
- [22] X. Sun, et al., A forest fire prediction model based on cellular automata and machine learning, *IEEE Access* 12 (2024) 55389–55403, <https://doi.org/10.1109/access.2024.3389035>.
- [23] D.D.B. Perrakis, et al., Improved logistic models of crown fire probability in Canadian conifer forests, *Int. J. Wildland Fire* 32 (10) (2023) 1455–1473, <https://doi.org/10.1071/wf23074>.
- [24] U. Oliveira, et al., A near real-time web-system for predicting fire spread across the Cerrado biome, *Sci. Rep.* 13 (1) (2023), <https://doi.org/10.1038/s41598-023-30560-9>.
- [25] Q.E. Barber, et al., The Canadian fire spread dataset, *Sci. Data* 11 (1) (2024), <https://doi.org/10.1038/s41597-024-03436-4>.
- [26] The family of ERA5 datasets - Copernicus Knowledge Base - ECMWF Confluence Wiki. <https://confluence.ecmwf.int/display/CKB/The,+family+of+ERA5+datasets?src=contextnavpagetreemode>.